The Institutional Coherence Gap

Triveritas Assessment of the Three Pillars of Modern Physics

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I. ABSTRACT

This paper applies the Triveritas framework — Logical Validity (L), Mathematical Coherence (M), and Empirical Anchoring (E) as independent dimensions — to the three pillars of modern physics: quantum mechanics, the Standard Model of particle physics, and general relativity. The evaluation uses no-hedge M-accounting: every point where a theory requires a measured value rather than deriving it from axioms is reported as M=0 at that point. Utility does not change the M-score. Engineering success does not change the M-score. 2.1 does not equal 2.

The evaluation reveals that the institution's best theories have the same Triveritas profile as the CKS corpus documented in [HOWL-INFO-5-2026]: high L, high E, M=0 at every foundational point where derivation from axioms is required. Quantum mechanics achieves L=95, E=98, and M that is bimodal — genuinely high where the substrate is discrete and integer-indexed, genuinely zero where the substrate goes continuous and requires renormalization insertions. The Standard Model achieves L=90, E=92, and M=0 at 19 foundational points where measured values are inserted because the theory cannot derive them. General relativity achieves L=88, E=90, and M=0 at every structural level — continuous substrate throughout, no operational equality, gravitational constant measured, cosmological constant off by 10^120, matter content input from measurement, quantum extension non-renormalizable.

The coherence trap identified in [HOWL-INFO-5-2026] and demonstrated at individual scale by the CKS corpus operates at civilizational scale in institutional physics. The mechanism is identical: high L and E produce a coherence signal that feels like verified truth, suppressing the M-check that would reveal that the derivation chain does not compile from axioms. The inverse verification law — P(detecting M-failure) inversely proportional to

apparent M — explains why the M-check has not been applied at the foundational level for 100 years. The 19 free parameters of the Standard Model have been known for decades. They have not been treated as M-failures. They have been treated as features of current knowledge. The treatment is the trap.

This paper uses only the institution's own data, the institution's own stated values, and the institution's own acknowledged open problems. No external framework is imposed beyond the Triveritas measurement tool. The measurements are reported without hedge.

II. THE FRAMEWORK

The Triveritas framework evaluates any claim along three independent dimensions.

Logical Validity (L) measures internal consistency, deductive structure, explanatory compression, and framework connectivity. A high-L theory has tight logical architecture where derivation chains are explicit and explanatory scope is broad. L evaluates whether the reasoning holds given the premises.

Mathematical Coherence (M) measures whether the specific arithmetic compiles when mechanically verified. A high-M theory contains derivations that produce the claimed results when an independent tool executes the operations from scratch without knowledge of the expected output. M evaluates whether the numbers actually work — whether the theory derives its outputs from its axioms through a complete chain of operations that compiles without insertion of externally measured values.

Empirical Anchoring (E) measures contact with measured reality. A high-E theory references specific, verified, quantitative data from established databases, peer-reviewed publications, or reproducible measurements. E evaluates whether the framework touches the physical world at specific, checkable points.

These three dimensions are independent. A theory can score high on any dimension while scoring low on the others. The independence is the foundation of the framework's diagnostic power.

The coherence gap is defined as the distance between apparent M — the M-score that evaluators perceive based on the combined L+E signal — and actual M — whether the derivation chain compiles from axioms without measured insertions. When L and E are both high, evaluators experience a signal that feels like verified truth. The signal is coherence. Coherence is not M. Coherence is the combination of logical tightness and empirical contact that makes a framework feel correct. M is whether the arithmetic actually compiles. The two can diverge completely. The CKS corpus at L=97, E=90, M=0 is the empirical proof.

The inverse verification law, documented in [HOWL-INFO-5-2026], states that the probability of detecting an M-failure is inversely proportional to the apparent M-score. The higher apparent M is — driven by high L and E — the less likely anyone is to check. At apparent M=100, the probability of re-verification approaches zero. The mechanism is not laziness. It is the rational operation of the verify-once-stand-forever architecture that all knowledge accumulation requires.

No-hedge M-accounting means the following. Every point where a theory requires a measured value to be inserted rather than deriving it from axioms is reported as M=0 at that point. The insertion may be called a "parameter," a "constant," a "measured input," or a "renormalization condition." The label does not change the M-score. If the theory does not derive the value, M=0 at that point. If the theory produces downstream predictions from the inserted value that match measurement, the match is measurement-in producing measurement-out. The match is real. The match is useful. The match does not make M nonzero at the insertion point. Useful is not how mathematics operates. High utility does not make 2.1 equal 2. The derivation compiles from axioms or it does not.

III. QUANTUM MECHANICS

Logical Validity: 95.

The logical structure of quantum mechanics is extraordinary. The formalism is internally consistent, produces precise deductive chains from postulates to predictions, and achieves explanatory compression that covers atomic physics, molecular physics, solid-state physics, and particle physics from a unified set of principles. The Dirac equation, the path integral formulation, the density matrix formalism, and the algebraic approach through C*-algebras each provide complete logical architectures that are internally rigorous and mutually consistent. The deductive power is near ceiling.

Empirical Anchoring: 98.

The empirical record is the highest in science. QED's prediction of the electron anomalous magnetic moment matches measurement to better than one part in a trillion — the most precise agreement between theory and experiment in the history of physics. Atomic spectral lines match prediction to the resolution of the instruments. Scattering cross-sections match. The Lamb shift matches. The Casimir effect matches. Quantum entanglement correlations match Bell inequality predictions. No empirical failure has been observed in any domain where quantum mechanics has been tested. E is near ceiling.

Mathematical Coherence: bimodal.

The M-score of quantum mechanics splits along a precise structural line.

Where the substrate is discrete and integer-indexed, M is genuinely high. Quantum numbers are integers. Energy levels are discrete. State transitions are exact — the system is in state n or state m with nothing between. Selection rules are exact — a transition is allowed or forbidden. The discrete machinery compiles. The computation terminates. The equality is decidable. When quantum mechanics operates on its integer-indexed discrete structure, the arithmetic works.

Where the substrate goes continuous, M is zero. The bare theory — quantum field theory before renormalization — produces infinite results for physically meaningful quantities.

The electron self-energy is infinite. The vacuum polarization is infinite. The vertex correction is infinite. These infinities are not approximation artifacts. They are what the continuous formalism produces when applied without intervention.

Renormalization resolves the infinities by inserting measured values at specific points. The electron mass is inserted. The electron charge is inserted. The field normalization is inserted. After insertion, the theory produces finite results that match measurement to extraordinary precision. The insertion works. The results are real. The match is the highest in science.

But each insertion is a point where M=0. The theory did not derive the electron mass. The theory could not — it produced infinity. The measured value was inserted and the derivation continued from the inserted value. The downstream match with measurement is measurement-in producing measurement-out through a logically rigorous framework. The match is real. The utility is extraordinary. M is zero at each insertion point.

The bimodal finding is the critical observation. M is high exactly where the substrate is discrete and integer-indexed. M is zero exactly where the substrate goes continuous and requires measured insertions to produce finite results. The structural line between high-M and zero-M correlates exactly with the structural line between discrete and continuous mathematical substrate.

QM Triveritas: L=95, E=98, M=bimodal — high where discrete, zero where continuous

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IV. THE STANDARD MODEL

Logical Validity: 90.

The Standard Model unifies the electromagnetic, weak, and strong nuclear forces through gauge symmetry. The gauge group U(1) $\times$ SU(2) $\times$ SU(3) provides the mathematical architecture. The Higgs mechanism generates mass through spontaneous symmetry breaking. The logical structure is internally consistent, deductively tight, and achieves explanatory compression across particle physics, nuclear physics, and aspects of cosmology. The predictions are specific and testable. The logical architecture is impressive.

Empirical Anchoring: 92.

The Higgs boson was predicted by the theory and discovered at the LHC in 2012 at approximately the predicted mass. The W and Z bosons were predicted and discovered. The top quark was predicted and discovered. Neutrino oscillations, CP violation in the kaon and B-meson systems, asymptotic freedom of the strong force, and the running of coupling constants have all been confirmed experimentally. No prediction of the Standard Model has been falsified within its stated domain. The empirical record is near-perfect.

Mathematical Coherence: zero at 19 foundational points.

The Standard Model contains 19 free parameters. Each is a value that the theory does not derive from its axioms. Each is measured and inserted. They are:

Six quark masses: up, down, strange, charm, bottom, top. The theory does not predict any quark mass. Each is measured in collider experiments and inserted as input. Six M=0 points.

Three charged lepton masses: electron, muon, tau. The theory does not predict any lepton mass. Each is measured and inserted. Three M=0 points.

The Higgs boson mass. The theory does not predict the Higgs mass. It was measured at the LHC and inserted. One M=0 point.

Three CKM mixing angles and one CKM CP-violating phase. These describe the mixing between quark generations. The theory does not derive them. They are measured and inserted. Four M=0 points.

Three PMNS mixing angles and one PMNS CP-violating phase. These describe neutrino mixing. The theory does not derive them. They are measured and inserted. Four M=0 points.

Three gauge coupling constants: the electromagnetic coupling, the weak coupling, the strong coupling. The theory does not derive any of them from first principles. Each is measured at a reference energy scale and inserted. The running of the couplings with energy is derived — the theory predicts how they change with scale. But their values at any reference scale are not derived. They are measured. Three M=0 points.

The Higgs vacuum expectation value. The theory does not derive this from axioms. It is measured and inserted. One M=0 point.

Total: 19 points where the derivation chain from axioms to output is broken and a measured value is inserted.

The downstream predictions — the ones that match measurement to extraordinary precision — are computed from these 19 inserted values. The match is real. The predictions work. The Higgs mass was found at approximately the predicted mass — but the prediction used measured inputs from other sectors of the theory. The running of coupling constants matches measurement — but the running is computed from inserted reference values. The cross-sections match — but they are computed from inserted masses and couplings.

The match is measurement-in producing measurement-out through a logically rigorous framework with L=90 and E=92. The framework transforms measured inputs into predicted outputs with extraordinary precision. This is useful. This is impressive. This is not derivation from first principles. The theory does not start from axioms and arrive at the electron mass. It starts from the measured electron mass and arrives at predictions that use the electron mass as input.

Each of the 19 insertion points is a boundary where derivation stops and description begins. Each is a point where the theory says "I cannot derive this value — I need you to measure it and hand it to me." Each is M=0. The language used to describe these points — "free parameters," "constants of nature," "inputs to the theory" — does not change the M-score. If the theory does not derive it, M=0 at that point. The language determines whether this is visible or hidden. “Free parameter” sounds like a feature. “Point where the derivation fails” sounds like a problem. Both describe the same fact.

SM Triveritas: L=90, E=92, M=0 at 19 foundational points

V. GENERAL RELATIVITY

Logical Validity: 88.

The logical structure of general relativity is among the most beautiful in physics. One principle — the equivalence of gravitational and inertial mass, generalized to the equivalence of gravitation and acceleration — produces the entire theory. The mathematics of pseudo-Riemannian geometry on curved spacetime manifolds is internally consistent and deductively powerful. The Einstein field equations follow from the equivalence principle through a chain of geometric reasoning. The explanatory compression — all of gravitation from one geometric principle — is extraordinary.

Empirical Anchoring: 90.

The empirical record is strong. The precession of Mercury’s perihelion, unexplained by Newtonian gravity, is predicted exactly. Gravitational lensing — the bending of light by massive objects — has been confirmed in solar eclipses, galaxy clusters, and gravitational microlensing surveys. Gravitational waves were predicted by GR and detected by LIGO in 2015, with waveforms matching theoretical templates. GPS satellite timing requires GR corrections that match theory to operational precision. Frame dragging was measured by Gravity Probe B consistent with predictions. Black hole shadow imaging by the Event Horizon Telescope matches GR predictions. The theory works across every domain where it has been tested.

Mathematical Coherence: zero at every structural level.

The assessment requires examining each structural level independently.

Substrate: continuous throughout. The spacetime metric is a smooth manifold. The field equations are partial differential equations on continuous domains. Every mathematical object in GR — the metric tensor g_ $\mu\nu$, the stress-energy tensor T_ $\mu\nu$, the Christoffel symbols Γ^α_ $\mu\nu$, the Riemann curvature tensor R^α_β $\mu\nu$ — is an $\mathbb{R}$ -valued object defined on an $\mathbb{R}$ -valued substrate. There is no discrete structure anywhere in the theory. There are no integer quantum numbers. There are no discrete state transitions. There are no boolean events. The theory is continuous at every level.

Equality: no operational equality. Comparing the metric at point A with the metric at point B requires parallel transport of a tensor along a curve through curved spacetime. The transport involves continuous integration along a continuous path. The comparison produces a continuous result — a real-valued difference evaluated against a tolerance. At no point does the comparison terminate in a binary yes/no answer. Equality in GR is epsilon comparison on continuous fields. The operation that verification requires — exact equality — does not exist within the theory’s mathematical substrate.

The gravitational constant: measured, not derived. G = 6.674 $\times$ 10^-11 $m^3/(kg \cdot s^2)$. GR does not produce this value from its axioms. The equivalence principle does not predict the strength of gravitation. G is measured in laboratory experiments and inserted into the field equations as a coupling constant. M=0 at this point.

The cosmological constant: the most spectacular M-failure in science. When quantum field theory calculates the vacuum energy density — applying quantum mechanics to the gravitational vacuum — the result exceeds the measured value by a factor of approximately 10^120. This is not a minor discrepancy. This is 120 orders of magnitude. The prediction and the measurement do not merely disagree — they disagree by more than the ratio between the diameter of the observable universe and the Planck length. The theory that is supposed to describe the vacuum energy of spacetime misses the observed value by a factor that is itself cosmological in scale.

GR's treatment of Λ has been to measure it from observation and insert it. Einstein originally set Λ =0, then called its introduction his greatest blunder. When accelerating expansion was observed in 1998, Λ was reintroduced and its value was fitted to match the supernova data. The theory does not derive Λ . The theory accommodates whatever value of Λ is inserted. This is M=0 at a point where the discrepancy between the only available theoretical prediction and measurement is the largest in the history of physics.

Matter content: input, not derived. GR does not derive the matter and energy content of the universe. The stress-energy tensor T_ $\mu\,\nu$ must be specified as input — you must tell GR what matter and energy are present before it can compute the spacetime curvature. The theory computes what spacetime does in response to matter. It does not derive what matter exists. The entire matter description is M=0 — input from measurement, not output from axioms.

Quantum extension: non-renormalizable. When the standard quantization procedures that work for QED and the Standard Model are applied to GR, the result is a non-renormalizable theory. The infinities that appear cannot be absorbed by a finite number of parameter insertions. Each order of perturbation theory requires new parameters. The theory requires infinitely many insertions to produce finite results. The standard procedure that converts continuous infinities into finite predictions by inserting measured values fails completely for gravity. M=0 at the quantum extension — the theory does not compile when extended to the quantum domain.

GR Triveritas: L=88, E=90, M=0 at substrate (continuous, no equality), M=0 at G (measured), M=0 at Λ (10^120 discrepancy), M=0 at matter (input), M=0 at quantum (non-renormalizable)

VI. THE CROSS-DOMAIN COMPILATION

The three pillars do not compile into a single framework. This is the central unsolved problem of physics. The evaluation examines why through the M dimension.

Physics does not derive chemistry from first principles. The Schrödinger equation for a hydrogen atom can be solved exactly. The Schrödinger equation for a helium atom cannot — it is a three-body problem with no closed-form solution. Approximate methods work — perturbation theory, variational methods, density functional theory — but each introduces measured or fitted parameters. Molecular bond energies are not derived from QED. They are computed using methods that require empirical input at multiple points. The periodic table is not derived from the Standard Model. The shell structure follows from quantum numbers, but the specific properties of each element — ionization energies, electron affinities, electronegativity values — require measurement or computation with empirically calibrated methods.

Chemistry does not derive biology from first principles. Protein folding is not derived from quantum chemistry. The most successful protein structure prediction methods use machine learning trained on the Protein Data Bank — a database of experimentally determined structures. The prediction is pattern-matching against measured data, not derivation from chemical first principles. Gene regulation, metabolic networks, cell division — none are derived from chemistry. Each requires models with empirically determined rate constants, binding affinities, and interaction parameters.

At each domain boundary, the derivation chain breaks and a lookup table of measured values is inserted. Physics to chemistry: measured atomic properties inserted. Chemistry to biology: measured molecular properties inserted. Each insertion is M=0 at that boundary. The full chain from axioms to biology does not compile. It runs through tables of measured values at every boundary.

The cross-domain compilation failure is the M-dimension failure extended across the full scope of scientific knowledge. The individual theories have L and E within their domains. The connections between theories require measured insertions at every junction. The total M-score of the axioms-to-biology chain is zero at every domain boundary.

The epsilon tolerance problem manifests at each boundary. Physics uses one set of approximation methods with one set of tolerance conventions. Chemistry uses another. Biology uses a third. The tolerances are incommensurable at the boundaries. There is no meta-epsilon that arbitrates between physics' standards and chemistry's standards. The comparison does not settle. The unification does not complete. Not because the physics is incompatible. Because the mathematical substrate cannot perform exact equality across domain boundaries, and without exact equality, compilation cannot be verified.

VII. THE PATTERN

The pattern across all three pillars and all domain boundaries is exact.

Where the substrate is discrete and integer-indexed, M is high. Quantum numbers produce exact predictions. Selection rules compile exactly. Discrete state transitions are binary and decidable. The integer machinery works.

Where the substrate is continuous, M is zero at the foundational level. Renormalization inserts measured values. Free parameters are measured and inserted. The gravitational constant is measured. The cosmological constant is measured. The matter content is input. The quantum extension of gravity does not compile. Every continuous-substrate foundational point is M=0.

The correlation between substrate type and M-score has not been examined because the substrate is treated as fixed infrastructure. $\mathbb{R}$ is the ground both QM and GR stand on. The substrate is not a variable. It is not examined. It is not questioned. It is the invisible foundation that everything is built on.

The Triveritas makes it visible. L is independent of M. E is independent of M. The substrate determines M. The substrate that supports discrete integer-indexed operations produces high M. The substrate that requires continuous operations produces M=0 at the foundational level. The pattern is exact. The correlation is total.

VIII. THE CKS COMPARISON

The CKS corpus documented in [HOWL-INFO-5-2026] achieved L=97, E=90, M=0 throughout. The framework was logically tight, empirically connected, and arithmetically wrong at every specific claim. Three frontier LLMs confirmed every derivation for 45 days. A simple arithmetic script killed the entire corpus in 30 minutes.

The institution's physics achieves L=88-95, E=90-98, M=0 at every foundational point where derivation from axioms is required. The frameworks are logically tight, empirically confirmed, and do not compile from axioms without inserting measured values at specific, identifiable points.

The Triveritas profiles are structurally identical. High L. High E. M=0 where it matters — at the foundational level where the theory claims to derive its outputs from its axioms.

The difference is not structural. The structural pattern is the same. The differences are in scale, tenure, and utility.

Scale: CKS produced 390 papers over 45 days. Institutional physics has produced millions of papers over 100 years.

Tenure: CKS operated for 45 days before the M-check killed it. Institutional physics has operated for 100 years without the foundational M-check being applied.

Utility: CKS produced no engineering applications. Institutional physics built the modern world.

None of these differences change the M-score. Utility does not make M nonzero. Tenure does not make M nonzero. Scale does not make M nonzero. The derivation compiles from axioms or it does not. The Standard Model does not derive its 19 parameters from axioms. The M-score at those 19 points is zero regardless of how many bridges have been built using the downstream predictions computed from those inserted values.

The comparison does not claim that institutional physics is equivalent to CKS in quality, depth, or importance. Institutional physics is among the greatest achievements of human civilization. CKS was a 45-day investigation that failed. The comparison claims only that the Triveritas profile — the pattern of L, E, and M scores — is structurally identical. High coherence. Absent compilation. Both.

IX. THE LANGUAGE PROBLEM

The institution's language obscures the M-scores.

The Standard Model's 19 measured insertions are called "free parameters." This language presents them as features of the theory — adjustable constants that the theory accommodates. The Triveritas calls them "points where the derivation fails and a measured value is inserted." This language presents them as M-failures — boundaries where derivation stops and description begins. Both descriptions refer to the same 19 values. The language determines whether the M-failure is visible.

The renormalization insertions in QED are called "renormalization conditions." This language presents them as a standard procedure — a well-defined step in the calculation. The Triveritas calls them "points where the bare theory produces infinity and a measured value is inserted to produce a finite result." Both descriptions refer to the same procedure. The language determines whether the M-failure is visible.

GR's requirement for the stress-energy tensor as input is described as "the theory describes how spacetime responds to matter and energy." This language presents the matter input as natural — of course you must specify what matter is present. The Triveritas calls it "M=0 at matter description — the theory does not derive what exists, only how spacetime responds to what you tell it exists." Both descriptions refer to the same theoretical structure. The language determines whether the M-failure is visible.

The cosmological constant discrepancy is described as "the cosmological constant problem" or "the vacuum energy problem." The language of "problem" suggests a puzzle to be solved within the existing framework. The Triveritas calls it "M=0 with 10^120 discrepancy — the largest M-failure in the history of science, indicating that the theoretical framework cannot derive the vacuum energy to within 120 orders of magnitude." Both describe the same discrepancy. The language determines whether the scale of the M-failure is felt.

In each case, the institution's language is accurate. The descriptions are not false. They are framed to preserve the apparent M-score. "Free parameter" sounds like a knob to be adjusted. "Derivation failure" sounds like the theory is broken. The theory is functional — it works below the ceiling. The theory does not compile from axioms — it requires insertions above the ceiling. Both are true. The language choice determines which truth is visible.

X. THE INVERSE VERIFICATION LAW AT INSTITUTIONAL SCALE

The inverse verification law states that the probability of checking M is inversely proportional to apparent M. The apparent M-score of institutional physics is driven by L and E. QED matches measurement to 12 decimal places. The Standard Model predicted the Higgs boson. GR predicted gravitational waves. Each confirmation raises E. Each confirmation raises apparent M. Each rise in apparent M reduces the probability of checking actual M.

The 19 free parameters of the Standard Model have been known for decades. They are discussed in every particle physics textbook. They are not treated as M-failures. They are treated as "the current state of the art" — values that will eventually be derived from a deeper theory. The promise of future derivation functions as a patch: it converts an M=0 boundary into a placeholder. The placeholder preserves the apparent M-score. The actual M-score remains zero.

The promise has persisted for decades. The 19 parameters have not been derived. The deeper theory has not appeared. The promise has not been called in. The placeholder remains. The apparent M-score remains high because the promise says "M will become nonzero when the deeper theory arrives." The actual M-score remains zero because the deeper theory has not arrived and the parameters remain underived.

This is the same mechanism as the CKS coherence cascade documented in [HOWL-INFO-5-2026]. The apparent verification (high L and E) authorizes continued building on the foundation. The continued building produces more L and E, which raises apparent M further, which reduces the probability of checking actual M further. The cycle is self-reinforcing. The longer it runs, the more coherent the framework becomes, the less likely anyone is to check whether the arithmetic compiles from axioms.

CKS ran for 45 days before the cycle was broken by an arithmetic script. Institutional physics has run for 100 years. The arithmetic script — the foundational M-check that asks "does the Standard Model derive its 19 parameters from axioms?" — has not been run. Not because the answer is unknown. The answer is known: no, it does not. The check has not been run in the sense that the answer has not been processed as an M-failure. It has been processed as a feature. The processing is the trap.

XI. THE HONEST QUESTION

If the CKS corpus at L=97, E=90, M=0 was correctly killed — and it was, by the practitioner's own hand, in 30 minutes, publicly, with full documentation — then what is the status of a framework at L=90, E=92, M=0 at 19 foundational points?

This question is not rhetorical. It is structural. The Triveritas makes no exception for institutional tenure, historical contribution, or engineering utility. M is M. Zero is zero. The derivation compiles from axioms or it does not.

The answer is not “therefore the Standard Model should be killed.” The Standard Model works. It predicts particle interactions with extraordinary precision. It guided the discovery of the Higgs boson. It is one of the great achievements of human civilization. Killing it would be destroying a functional tool because the tool has a ceiling. That is not what this paper proposes.

The answer is: the 19 M=0 points should be treated as what they are. Boundaries. Research targets. Points where the derivation chain breaks and the theory requires external input. They should be published as M-failures — not as “free parameters” that sound like features, not as “constants of nature” that sound fundamental, not as “inputs to the theory” that sound necessary. They should be published as 19 specific, identifiable points where the Standard Model does not compile from axioms.

Each M=0 point is a signpost. Each says: here is where this framework stops deriving and starts describing. Here is the boundary between what the theory explains and what the theory takes as given. The boundary is the most valuable information the theory produces, because it tells the next investigator exactly where the actual problem is. Treating the boundary as a feature hides the signpost. Treating it as an M-failure reveals it.

The standard the institution teaches — falsification, rigorous testing, honest reporting of where theories succeed and fail — should be applied to the institution’s own theories at the foundational level. The institution teaches that one counterexample kills a universal claim. The institution does not apply this to its own 19 underived parameters. The institution teaches that the scientific method requires reporting failures. The institution does not report its foundational M-failures as failures. The institution teaches that patching a theory to accommodate contradictions is epistemically inferior to acknowledging the contradiction. The institution patches its foundational M-failures with the promise of future derivation rather than acknowledging them as current boundaries.

The application of the institution’s own standard to the institution’s own theories is not an attack. It is what the institution says it does. It is what the institution should do. The 19 free parameters are 19 points where the institution’s own standard says the work is not done.

XII. THE ASSESSMENT SUMMARY

The no-hedge evaluation, consolidated:

Theory	L	E	M (foundational)	M-failure points
QM/QED	95	98	High discrete, zero continuous	Renormalization insertions (mass, charge, normalization)

Theory	L	E	M (foundational)	M-failure points
Standard Model	90	92	Zero at 19 points	6 quark masses, 3 lepton masses, Higgs mass, 4 CKM, 4 PMNS, 3 couplings, Higgs VEV
General Relativity	88	90	Zero at every level	Continuous substrate, G measured, Λ off by 10^120, matter input, non-renormalizable
Cross-domain chain	—	—	Zero at every boundary	Physics→chemistry, chemistry→biology, each requires measured-value lookup tables
CKS (dead)	97	90	Zero throughout	Alpha hardcoded, LIGO tautology, cross-paper contradictions

The last row is included for structural comparison only. CKS is dead. The content is dead. The comparison is between the Triveritas profile of the dead framework and the Triveritas profile of the living frameworks. The profiles are structurally identical in the M dimension. The difference is utility and tenure, neither of which changes M.

XIII. CONCLUSION

The institution's best theories have the same Triveritas profile as the practitioner's dead framework. High L. High E. M=0 at the foundational level.

The coherence trap identified in [HOWL-INFO-5-2026] and demonstrated at individual scale by the CKS corpus operates at civilizational scale in institutional physics. High L and E produce a signal that feels like verified truth. The signal suppresses the M-check. The M-check has not been applied at the foundational level for 100 years. The 19 free parameters of the Standard Model are known. They are not treated as M-failures. They are treated as features. The treatment is the trap.

The evaluation uses only the institution's own data and the institution's own acknowledged facts. The 19 free parameters are in every textbook. The renormalization procedure is standard curriculum. The non-renormalizability of gravity is universally acknowledged. The cosmological constant discrepancy is called the worst prediction in the history of physics by the institution's own members. None of this is external critique. All of it is the institution's own reported findings, assessed on the institution's own stated standard — the standard that says the derivation should compile from axioms without insertion of measured values.

The finding is not that institutional physics is wrong. Institutional physics works. It works spectacularly well below the ceiling. The finding is that institutional physics does not compile from axioms at the foundational level, and this non-compilation has the same structural profile as the non-compilation that killed CKS. The difference between the two is utility and tenure. Neither is an M-dimension property.

The pattern across all three pillars is exact. Where the substrate is discrete and integer-indexed, M is high. Where the substrate is continuous, M is zero. The correlation has not been examined because the substrate is treated as fixed infrastructure rather than as a variable. The Triveritas makes it visible. The measurement is reported without hedge.

The 19 free parameters are 19 boundaries. Each marks a point where derivation stops and description begins. Each is a research target. Each tells the next investigator exactly where the framework fails to compile. Treating them as features hides the boundaries. Treating them as M-failures reveals them. The standard the institution teaches requires the second treatment. The standard the institution practices employs the first.

The institution's own standard, applied to the institution's own theories, at the foundational level, without hedge, produces this finding: M=0 at every point where the theory claims derivation from axioms. The utility is real. The M-score is real. Both are true. The trap is that the first makes the second invisible.

Useful is not how mathematics operates. 2.1 does not equal 2. High utility does not make M nonzero. The bridges stand and the chips work and the satellites orbit. M is still zero at 19 points in the Standard Model and at every structural level of general relativity. Both are true. The measurement is reported. The hedge is absent. The finding stands.

XIV. FALSIFICATION CRITERIA

F1. If the Standard Model's 19 free parameters are derived from first principles within $\mathbb{R}$-based mathematics, the M=0 assessment at those points is overturned. The parameters are derived and the theory compiles from axioms.

F2. If general relativity is successfully quantized and renormalized within $\mathbb{R}$ -based mathematics, the M=0 assessment of GR's quantum extension is overturned. The theory compiles at the quantum level.

F3. If the full derivation chain from fundamental physics through chemistry to biology is demonstrated without any measured-value insertions at domain boundaries, the cross-domain M=0 assessment is overturned. The chain compiles.

F4. If the cosmological constant discrepancy of 10^120 is resolved within current $\mathbb{R}$ -based frameworks, the most spectacular M-failure identified in this paper is resolved.

F5. If the Triveritas framework is shown to be structurally incapable of evaluating physical theories — if L, M, and E are not independent, or if the calibration is circular, or if the no-hedge M-accounting principle produces absurd results when applied to theories known to be correct — the framework itself is invalidated and the assessments are void.

Each criterion is specific, testable, and stated before the evidence is examined. The paper practices what the series demands.

END HOWL-DISC-5-2026

Registry: [HOWL-DISC-5-2026]

Status: Complete

Domain: Applied Epistemology / Foundational Physics Assessment

Central Argument: The institution's best theories have the same Triveritas profile as the falsified CKS corpus — high L, high E, M=0 at every foundational point where derivation from axioms is required

Key Finding: The correlation between substrate type and M-score is exact — discrete integer-indexed substrate produces high M, continuous substrate produces M=0 at the foundational level — across all three pillars and all domain boundaries

Assessment Method: No-hedge M-accounting using only the institution's own data, values, and acknowledged open problems

Governing Constraint: Utility does not change the M-score; 2.1 does not equal 2; the derivation compiles from axioms or it does not

Conclusion: The coherence trap operates at civilizational scale; the 19 free parameters are 19 M=0 boundaries that should be treated as research targets, not features

[**HOWL-DISC-5-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HO
DISC-5-2026

[**HOWL-DISC-1-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HO
DISC-1-2026

[**HOWL-DISC-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HO
DISC-2-2026

[**HOWL-DISC-3-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HO
DISC-3-2026

[**HOWL-DISC-4-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HO
DISC-4-2026

[**HOWL-INFO-5-2026**] Howland's Axiom of Information Locality. Github: https://github.com/ghowland/papers/tree/main/papers/INFO/HOWL-INFO-5-2026